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EXP 19 Implementation of Naive Bayes classification for Bank Loan prediction.

CODING:

```
import pandas as pd

from sklearn.model_selection import train_test_split from

sklearn.preprocessing import LabelEncoder from

sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report data = {

  'Age': [25, 30, 35, 40, 45, 28, 32, 38, 50, 29,

        42, 36, 27, 48, 34, 31, 44, 39, 26, 52],

  'Income': [25000, 35000, 45000, 60000, 75000, 30000, 40000, 55000,

          80000, 32000, 65000, 48000, 28000, 85000, 42000, 38000,

          70000, 58000, 27000, 90000],

  'LoanAmount': [100000, 150000, 200000, 250000, 300000, 120000,

             180000, 220000, 350000, 140000, 280000, 210000,

             110000, 320000, 190000, 160000, 270000, 230000,

             100000, 400000],

  'CreditScore': [650, 680, 700, 750, 780, 660, 690, 720, 800, 670,

              760, 710, 640, 790, 700, 680, 770, 730, 630, 810],

  'LoanStatus': [0, 0, 1, 1, 1, 0, 1, 1, 1, 0,

             1, 1, 0, 1, 1, 0, 1, 1, 0, 1]

}

df = pd.DataFrame(data)

print("Bank Loan Dataset")

print(" --------------------")

print(df)

X = df[['Age', 'Income', 'LoanAmount', 'CreditScore']] y =

df['LoanStatus']
```

```
X_train, X_test, y_train, y_test = train_test_split( X, y,
    test_size=0.30,
    random_state=42,
    stratify=y)
model = GaussianNB()
model.fit(X_train, y_train) y_pred =
model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("\nNaive Bayes Classification Results")
print("------------------------------------------")
print("Actual Values   :", y_test.values)
print("Predicted Values:", y_pred)
print("Accuracy        :", accuracy)
print("Accuracy (%)   :", accuracy * 100)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nClassification Report:")
print(classification_report(
    y_test, y_pred,
    target_names=['Loan Rejected', 'Loan Approved']))
new_applicant = [[35, 50000, 180000, 720]] prediction =
model.predict(new_applicant) print("\nNew Applicant
Details")
print("------------------------")
print("Age         :", new_applicant[0][0])
print("Income       :", new_applicant[0][1])
print("Loan Amount :", new_applicant[0][2])
```

```
print("Credit Score:", new_applicant[0][3]) if
prediction[0] == 1:
  print("Loan Prediction: APPROVED")
else:
  print("Loan Prediction: REJECTED")
```

OUTPUT:

```
Bank Loan Dataset
-----------------
    Age  Income  LoanAmount  CreditScore  LoanStatus
0    25   25000      100000          650           0
1    30   35000      150000          680           0
2    35   45000      200000          700           1
3    40   60000      250000          750           1
4    45   75000      300000          780           1
5    28   30000      120000          660           0
6    32   40000      180000          690           1
7    38   55000      220000          720           1
8    50   80000      350000          800           1
9    29   32000      140000          670           0
10   42   65000      280000          760           1
11   36   48000      210000          710           1
12   27   28000      110000          640           0
13   48   85000      320000          790           1
14   34   42000      190000          700           1
15   31   38000      160000          680           0
16   44   70000      270000          770           1
17   39   58000      230000          730           1
18   26   27000      100000          630           0
19   52   90000      400000          810           1

Naive Bayes Classification Results
-----------------------------------
Actual Values   : [0 1 0 1 1 1]
Predicted Values: [0 1 0 1 1 1]
Accuracy        : 1.0
Accuracy (%)    : 100.0

Confusion Matrix:
[[2 0]
 [0 4]]

Classification Report:
               precision    recall  f1-score   support

Loan Rejected       1.00      1.00      1.00         2
Loan Approved       1.00      1.00      1.00         4

     accuracy                           1.00         6
    macro avg       1.00      1.00      1.00         6
 weighted avg       1.00      1.00      1.00         6
```

```
New Applicant Details
---------------------
Age          : 35
Income       : 50000
Loan Amount : 180000
Credit Score: 720
Loan Prediction: APPROVED
```